ASSIGNMENT

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SUBMITTED TO:

DR. KAMAL MURARI

SUBMITTED BY:

GROUP 1

ADITI KISHORE M2024ECCSS001

ANASUA MUKHOPADHYAY M2024ECSS002

ANSHUMAN SINGH M2024ECSSS003

DUGUTA SREEJA M2024ECCSS004

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1. Sampling Design

1.1. RANDOM SAMPLING

Table 1. Population for Random Sampling

S. No.	HH ID	Name of the Head of Household	Age of HoH	Gender M=1 F=2	Area	Total Members	No. of years of Education of HoH	Primary Occupation of HoH Code (1-10)	No. of years of Education
1	A020	RAJANAND PANDAGARE	40	1	RAMABAI NAGAR	13	12	10	12
2	C086	SATISH	30	1	PHULE NAGAR	4	10	2	10
3	C107	SUMAN SUDAMA MAGHADE	65	1	PHULE NAGAR	3	0	1	0
4	B097	V.D. JANGLE	66	1	GARIB NAGAR	3	8	10	8
5	B096	PRAKASH INGLE	52	1	GARIB NAGAR	4	12	2	12
6	C102	VIDYADHAR	32	1	PHULE NAGAR	5	10	2	10
7	C103	TUKARAM SHEKHBA INGLE	65	1	PHULE NAGAR	5	7	1	7
8	B088	SUSHANT	37	1	GARIB NAGAR	2	9	3	9
9	B093	ZAREENA QURESHI	40	2	GARIB NAGAR	3	0	3	0
10	B094	PRASHANT MAGAR	35	1	GARIB NAGAR	3	10	3	10
11	C060	SHYAM BIHARI MAURYA	51	1	PHULE NAGAR	6	15	2	15
12	C061	GIRISH RAI	30	1	PHULE NAGAR	7	17	2	17
13	C062	URMILA DEVI	52	2	PHULE NAGAR	6	0	1	0
14	C063	BRIHASPAT CHAUDHARY	55	1	PHULE NAGAR	8	8	4	8
15	C065	SHOBHA	40	2	PHULE NAGAR	3	0	2	0
16	C077	KESHARBAI SUKHDEV PATEKAR	65	2	PHULE NAGAR	1	0	10	0
17	C079	VILAS UTTAM	42	1	PHULE NAGAR	4	7	4	7
18	C080	SUNDAR BHISHMA JAMRAO	40	1	PHULE NAGAR	5	15	3	15
19	C081	MADHUKAR SAWANT	48	1	PHULE NAGAR	5	11	4	11
20	C082	SUNIL SAWANT	40	1	PHULE NAGAR	6	6	9	6
21	C083	VIJAYA BARU	55	2	PHULE NAGAR	6	0	2	0
22	C084	CRISTU RAJA	44	1	PHULE NAGAR	5	8	2	8
23	B045	MANOJ KUMAR	23	1	GARIB NAGAR	6	15	3	15
24	B038	SANJAY SHINDE	26	1	GARIB NAGAR	7	12	3	12

25	B035	SANTOSH DHOTRE	32	1	GARIB NAGAR	4	10	3	10
26	A015	PADMAVATI SHIRSAT	50	2	RAMABAI NAGAR	5	0	1	0
27	B037	ARUN KAMBLE	54	1	GARIB NAGAR	2	10	10	10
28	B047	RAVI BHIMRAO KHAIRAR	42	1	GARIB NAGAR	4	8	3	8
29	B049	PUSHPA DEVI VISHWAKARMA	55	2	GARIB NAGAR	2	0	1	0
30	B050	MANGESH	43	1	GAREEB NAGAR	5	12	2	12
31	B051	RESHMA	35	2	GAREEB NAGAR	4	10	2	10
32	B053	KUNDLI GHUGHE	50	1	GAREEB NAGAR	7	0	3	0
33	B006	VINOD	35	1	GAREEB NAGAR	3	10	2	10
34	C034	RAVI UMAJI PANDIT	34	1	PHULE NAGAR	4	10	2	10
35	C032	NANDLAL GUPTA	60	1	PHULE NAGAR	3	8	4	8
36	C030	ASHOK B. JANRAO	68	1	PHULE NAGAR	10	7	6	7
37	C028	JITENDRA	35	1	PHULE NAGAR	5	6	2	6
38	C025	PRABHAKAR MORE	66	1	PHULE NAGAR	7	4	10	4
39	C022	VIJAY BAHADUR MISHRA	60	1	PHULE NAGAR	7	12	9	12
40	C021	RAVI KISHOR	30	1	PHULE NAGAR	7	10	3	10
41	C004	BABU FAKIRAO GAEKWAD	46	1	PHULE NAGAR	6	5	2	5
42	C026	BALASAHEB GAJDHONE	62	1	PHULE NAGAR	5	12	2	12
43	C008	RANU NARAYAN AGALE	38	1	PHULE NAGAR	5	10	3	10
44	C003	DADARAO	40	1	PHULE NAGAR	5	10	2	10
45	A007	VITHAL LAXMAN WAGHMARE	40	1	RAMABAI NAGAR	3	12	2	12
46	A008	GAURAV SRI SART	40	1	RAMABAI NAGAR	4	8	4	8
47	A009	GOPAL MATHUR	32	1	RAMABAI NAGAR	3	12	2	12
48	B062	ARJUN NIHINAT	31	1	GAREEB NAGAR	2	12	3	12
49	B060	MANI	50	2	GAREEB NAGAR	3	0	2	0
50	B059	CHANDRAKANT	65	1	GAREEB NAGAR	4	4	10	4
51	B057	SANTOSH	38	1	GAREEB NAGAR	3	12	3	12
52	B001	SUDAM RAJ GURU	50	1	GARIB NAGAR	3	0	3	0
53	B002	BABBAN YADAV	50	1	GARIB NAGAR	4	10	3	10
54	B003	ANURADHA	40	2	GARIB NAGAR	10	2	2	2
55	B005	PANDURANG NAIK	32	1	GARIB NAGAR	6	12	3	12
56	B010	NANDAKISHORE TATERAO BORDE	28	1	GARIB NAGAR	3	10	2	10

57	B011	MILIND	34	1	GARIB NAGAR	4	12	2	12
58	B012	MIRABAI SHINDE	65	2	GARIB NAGAR	1	4	4	4
59	B013	SAVITA SIROHI	33	2	GARIB NAGAR	4	7	3	7
60	B014	DHARMENDRA VISHWAKARMA	35	1	GARIB NAGAR	5	10	4	10
61	B015	ABDUL HAMID	60	1	GARIB NAGAR	2	0	10	0
62	B016	KAMLA YADAV	60	1	GARIB NAGAR	5	10	2	10
63	B017	INAMULAH KHAN	30	1	GARIB NAGAR	4	7	2	7
64	B018	HARISH SAWADEKAR	30	1	GARIB NAGAR	2	11	3	11
65	C009	MACHHINDRA WAGHMARE	32	1	PHULE NAGAR	6	15	2	15
66	C010	LAXMAN AGALE	55	1	PHULE NAGAR	7	5	2	5
67	A012	PREM NARAYAN PANDEY	52	1	RAMABAI AMBEDKAR NAGAR	2	15	3	15
68	A016	PADMAVATI PANDEY	50	2	RAMABAI AMBEDKAR NAGAR	6	7	1	7
69	B036	RAM BARAN	60	1	GARIB NAGAR	5	7	10	7
70	B063	SM SHARMA	55	1	GARIB NAGAR	2	8	3	8
71	B064	JAYSHREE NIKAM	33	2	GARIB NAGAR	3	8	3	8
72	B065	BABROO SHAMRAO GADE	50	1	GARIB NAGAR	5	10	3	10
73	B066	SANJAY JADHAV	33	1	GARIB NAGAR	4	9	3	9
74	B068	SHYAM LAL PAL	51	1	GARIB NAGAR	7	7	2	7
75	B069	KAVITA SOMIONKAR	47	2	GARIB NAGAR	3	10	3	10
76	B071	SR YADAV	58	1	GARIB NAGAR	7	5	3	5
77	B072	GAYA NAMDEO ADAO	50	2	GARIB NAGAR	1	4	2	4
78	B073	JALINDER UBALE	57	1	GARIB NAGAR	11	4	3	4
79	B074	DASHRAT KAMBLE	68	1	GARIB NAGAR	8	0	4	0
80	B075	Md. SALIM	50	1	GARIB NAGAR	5	0	4	0
81	B076	NANDA KAKDE	48	2	GARIB NAGAR	2	4	2	4
82	B078	RAMCHANDRA MURDAM	36	1	GARIB NAGAR	6	10	2	10
83	B085	MANOJ KUMAR UPADHYAYA	33	1	GARIB NAGAR	4	9	4	9
84	B089	HARI RAM	39	1	GARIB NAGAR	10	9	4	9
85	B092	SHEIKH JAHANGIR	65	1	GARIB NAGAR	3	0	4	0
86	C035	AMIT SALVE	59	1	PHULE NAGAR	4	3	2	3
87	C039	DHARAMPAL NADAR	58	1	PHULE NAGAR	3	5	2	5

88	C042	DAYACHAND MAURYA	57	1	PHULE NAGAR	5	9	3	9
89	C045	LAXMAN NARAYAN BHADARGE	45	2	PHULE NAGAR	5	0	2	0
90	C048	PREMCHAND NISHAD	50	1	PHULE NAGAR	6	0	2	0
91	C051	NEELA PHADKE	40	2	PHULE NAGAR	4	0	4	0
92	C053	BALDEEP NIRALI	59	1	PHULE NAGAR	5	0	3	0
93	C055	GORAKHNATH LOKHANDE	60	1	PHULE NAGAR	7	2	9	2
94	C057	ARUN CHAMATHA	59	1	PHULE NAGAR	4	0	2	0
95	C089	MALATI	50	2	PHULE NAGAR	3	7	2	7
96	C059	PRADEEP CHAWAN	55	1	PHULE NAGAR	3	5	2	5
97	A018	SUSHILA ASHOK JADHAV	54	2	RAMABAI AMBEDKAR NAGAR	4	7	2	7
98	C092	VINOD ASHOK INGLE	32	1	PHULE NAGAR	3	10	2	10
99	C095	DHAMMA PAL	29	1	PHULE NAGAR	4	10	2	10
100	C097	KAUSHALYA GANDHLE	60	2	PHULE NAGAR	4	3	2	3
101	C098	CHANDRAKANT GADE	40	1	PHULE NAGAR	4	10	2	10
102	C099	UMESH SHINDE	30	1	PHULE NAGAR	2	12	6	12
103	C100	NATHA ROKADE	39	1	PHULE NAGAR	2	9	2	9
104	A002	Mira Dubey	45	2	RAMABAI AMBEDKARNAGAR	6	5	6	5
105	A003	Kamalabai	70	2	RAMABAI AMBEDKARNAGAR	8	0	1	0
106	A010	Prakash	47	1	RAMABAI AMBEDKARNAGAR	4	7	2	7
107	C006	Mahesh Kamble	25	1	PHULENAGAR	2	12	2	12
108	C013	Anita Singh	42	2	PHULENAGAR	5	8	1	8
109	C017	Urmila Vishwakarma	45	2	PHULENAGAR	7	8	1	8
110	C020	Premshila	40	2	PHULENAGAR	5	10	4	10
111	C023	Sukhdev Sawant	56	1	PHULENAGAR	5	12	3	12
112	A014	Muralidhar	55	1	RAMABAI AMBEDKARNAGAR	5	10	3	10
113	B025	Noorjahan	40	2	GARIBNAGAR	5	0	1	0
114	B028	Rahibai Kewde	60	2	GARIBNAGAR	7	4	2	4
115	B031	Mangal	32	1	GARIBNAGAR	5	7	2	7
116	B041	Shanti Singh	35	2	GARIBNAGAR	5	12	1	12
117	B044	Ponraj Nadar	47	1	GARIBNAGAR	4	8	4	8
118	B055	Rajendra Raikwar	31	1	GARIBNAGAR	4	10	2	10
119	B056	Sandeep	32	1	GARIBNAGAR	5	12	2	12

120	B086	Naresh Khandu	34	1	GARIBNAGAR	4	4	2	4
121	B091	Nasir Ahmed Sheikh	70	1	GARIBNAGAR	5	0	6	0
122	C036	Ashutosh Gade	58	1	GARIBNAGAR	4	3	2	3
123	C038	Ankita Wahude	55	2	PHULENAGAR	9	3	3	3
124	C041	Raju Solanki	45	1	PHULENAGAR	3	8	3	8
125	C044	Kamal Jadhav	40	1	PHULENAGAR	4	0	2	0
126	C047	Kanchan	23	2	PHULENAGAR	5	10	3	10
127	C050	Vinay Shinde	32	1	PHULENAGAR	3	10	3	10
128	A021	Ramesh Pandagare	49	1	RAMABAI AMBEDKARNAGAR	4	8	3	8
129	A022	Kiran Dhediya	41	1	RAMABAI AMBEDKARNAGAR	5	12	4	12
130	A023	Govind Sahni	35	1	RAMABAI AMBEDKARNAGAR	6	10	2	10
131	C093	Seema	32	2	PHULENAGAR	6	7	2	7
132	C096	Sugandh Rokade	37	1	PHULENAGAR	3	8	2	8
133	C108	Nandeo Pandit	55	1	PHULENAGAR	6	9	2	9
134	C109	Mohammed Sulaiman Dhulji Chauhan	39	1	PHULENAGAR	7	8	4	8
135	B008	Babasaheb	42	1	GARIB NAGAR	4	0	2	0
136	B007	Md. Asgar Naddaf	31	1	GARIB NAGAR	4	7	2	7
137	C033	Bandu Pandit	35	1	PHULE NAGAR	8	15	10	15
138	C031	Santosh Patil	28	1	PHULE NAGAR	5	10	3	10
139	C029	Kamal Ramdev Gawai	50	2	PHULE NAGAR	6	2	1	2
140	C027	Sukhdev Jadhav	50	1	PHULE NAGAR	5	0	2	0
141	C024	Mahadev Hazare	58	1	PHULE NAGAR	4	0	3	0
142	C019	Brijesh Kumar Pandey	34	1	PHULE NAGAR	9	12	4	12
143	C016	Pradeep Ashok Godhera	26	1	PHULE NAGAR	5	11	2	11
144	C005	Ambadas Bhimrao Nimbadkar	63	1	PHULE NAGAR	7	5	10	5
145	A011	Sreedhar Vishwanath Kopalwale	40	1	RAMABAI AMBEDKAR NAGAR	4	15	3	15
146	A006	Salubai Ramdas Ayera	60	2	RAMABAI AMBEDKAR NAGAR	10	0	1	0
147	A004	Hasan Raikbar	51	1	RAMABAI AMBEDKAR NAGAR	8	3	2	3
148	B024	Md. Azad Khan	45	1	GARIB NAGAR	9	0	4	0
149	A013	Manohar P.	78	1	RAMABAI AMBEDKAR NAGAR	9	4	6	4

150	B067	Ramlochan Maurya	52	1	GARIB NAGAR	8	10	4	10
151	B070	Dayashankar Pandey	68	1	GARIB NAGAR	6	5	6	5
152	B061	Vinod Gambre	30	1	GARIB NAGAR	4	30	10	30
153	B039	Ranjit Rathore	34	1	GARIB NAGAR	5	10	2	10
154	B033	Rajesh Trimbak Nikajhe	40	1	GARIB NAGAR	5	7	2	7
155	B030	Abdul Rahim Sheikh	48	1	GARIB NAGAR	5	4	4	4
156	B054	Shoaib Shaikh	35	1	GARIB NAGAR	5	0	4	0
157	B046	Kalyani	47	2	GARIB NAGAR	4	0	5	0
158	B042	Aakash Chauhan	45	1	GARIB NAGAR	4	10	2	10
159	B048	Subadra	56	2	GARIB NAGAR	4	0	2	0
160	B052	Rajendra Jadhav	53	1	GARIB NAGAR	4	5	4	5
161	B058	Vijay	33	1	GARIB NAGAR	3	10	2	10
162	B027	Dhanwantari Parve	37	2	GARIB NAGAR	3	7	2	7
163	C043	Satyadev Rajbhar	55	1	PHULE NAGAR	11	9	3	9
164	C040	Dilip Padiyar	61	1	PHULE NAGAR	5	3	2	3
165	B077	Satish Jadav	47	1	GARIB NAGAR	5	9	2	9
166	C075	Chandu Chalke	45	1	PHULE NAGAR	5	0	2	0
167	C056	Ganesh Bhadargre	50	1	PHULE NAGAR	5	2	6	2
168	C068	Hasina Sheikh	30	2	PHULE NAGAR	5	0	1	0
169	C070	Kamal	46	2	PHULE NAGAR	5	0	2	0
170	C076	Monica Shantanu Shinde	36	2	PHULE NAGAR	4	9	2	9
171	C085	Siddhart Chauhan	30	1	PHULE NAGAR	4	12	2	12
172	C058	Damodar Koli	52	1	PHULE NAGAR	4	4	2	4
173	C049	Dukni Nishad Malah	70	2	PHULE NAGAR	4	0	1	0
174	C037	Arjun Shinde	54	1	PHULE NAGAR	4	9	3	9
175	C046	Santosh Sasare	28	1	PHULE NAGAR	3	7	3	7
176	C052	Prem Shankar	56	1	PHULE NAGAR	3	0	3	0
177	C054	Santiam Mishra	56	1	PHULE NAGAR	3	3	2	3
178	C088	Nagarbai Hazare	80	2	PHULE NAGAR	7	0	1	0
179	A025	Rajesh Bimajhi	33	1	RAMABAI AMBEDKAR NAGAR	6	10	2	10
180	B100	Kijurao	51	1	GARIB NAGAR	5	0	2	0
181	B099	Rohidas Devgunubale	60	1	GARIB NAGAR	5	3	2	3

182	B098	Kailash Madhukar Tagatode	45	1	GARIB NAGAR	4	10	2	10
183	B101	Dinkar Nikalje	58	1	GARIB NAGAR	4	9	6	9
184	C087	Daya Bai	50	2	PHULE NAGAR	3	0	2	0
185	A005	Himmat	28	1	RAMABAI AMBEDKARNAGAR	3	10	4	10
186	B004	Mahadesh Ishwar Rathore	41	1	GARIBNAGAR	7	10	2	10
187	B005	Kalyan	35	2	GARIBNAGAR	5	0	2	0
188	B019	Naunat Kekan	52	1	GARIBNAGAR	3	15	3	15
189	B020	Kripa Shankar Pandey	45	1	GARIBNAGAR	4	0	3	0
190	B021	Shawantabai Sitaram	60	2	GARIBNAGAR	1	60	2	60
191	B022	Ramesh Sable	55	1	GARIBNAGAR	5	6	6	6
192	B023	Sarjahan	32	2	GARIBNAGAR	8	5	1	5
193	C001	Sandhya Zagade	45	2	PHULENAGAR	4	7	2	7
194	C002	Bhola Verma	45	1	PHULENAGAR	3	9	2	9
195	C007	Anil Sath	30	2	PHULENAGAR	4	6	2	6
196	C011	Shahtullah Sheikh	45	1	PHULENAGAR	5	7	2	7
197	C012	Harishchandra Maurya	60	1	PHULENAGAR	8	13	4	13
198	C015	Thangararaj	48	1	PHULENAGAR	8	0	2	0
199	C018	Amar Yadav	34	1	PHULENAGAR	4	17	3	17
200	A001	Satish Vinkar	36	1	RAMABAI AMBEDKARNAGAR	7	10	10	10
201	A017	Prabash	39	1	RAMABAI AMBEDKARNAGAR	4	17	3	17
202	B026	Asha Krishna Toke	38	2	GARIBNAGAR	2	0	2	0
203	B029	Vimal Pandagale	50	2	GARIBNAGAR	5	0	1	0
204	B032	Sunita Kishore Daudve	40	2	GARIBNAGAR	4	0	2	0
205	B034	Sunil Trimbak Nikaje	45	1	GARIBNAGAR	5	9	3	9
206	B040	Hari Gadge	47	1	GARIBNAGAR	4	7	2	7
207	B043	J.R. Mishra	64	1	GARIBNAGAR	5	8	6	8
208	B079	Surekha	40	2	GARIBNAGAR	7	0	1	0
209	B080	Dhani	26	2	GARIBNAGAR	1	0	1	0
210	B081	Santosh Jadhav	40	2	GARIBNAGAR	7	8	2	8
211	B082	Trimbal Dabhade	69	1	GARIBNAGAR	10	10	2	10
212	B083	Bhuvneshwar Pathak	60	1	GARIBNAGAR	9	10	10	10
213	B084	Shantaram	50	1	GARIBNAGAR	4	10	2	10

214	B087	Ramjitan Prajapati	55	1	GARIBNAGAR	5	10	2	10
215	B090	Subhan Sheikh	36	1	GARIBNAGAR	4	12	2	12
216	B095	Leela	65	2	GARIBNAGAR	4	0	2	0
217	C014	Sushila Devi Hazam	60	2	PHULENAGAR	8	0	1	0
218	C064	Ramashrai Pal	48	1	PHULENAGAR	2	8	2	8
219	C066	Sundarbai	45	2	PHULENAGAR	10	0	2	0
220	C067	Santa Babu Methades	55	2	PHULENAGAR	7	0	2	0
221	C069	Shantaram Sakaram Joshi	45	1	PHULENAGAR	4	15	4	15
222	C071	Salma Sheikh	50	2	PHULENAGAR	8	0	1	0
223	C072	Suman Yadav	38	2	PHULENAGAR	5	4	1	4
224	C073	Vithal Chandrapa Kariagowd	36	1	PHULENAGAR	5	7	4	7
225	C074	Praveen Saude	30	1	PHULENAGAR	5	9	4	9
226	C078	Bala Devi	58	2	PHULENAGAR	7	5	1	5
227	C090	Ansaruddin Sheikh	36	1	PHULENAGAR	10	12	4	12
228	C091	Arun Sikaram	35	1	PHULENAGAR	4	12	3	12
229	A019	Aasha	30	2	RAMABAI AMBEDKARNAGAR	5	7	2	7
230	A024	Sanjay Vinkar	40	1	RAMABAI AMBEDKARNAGAR	4	10	3	10
231	C094	Anand Sopah Pargaonbar	39	1	PHULENAGAR	4	10	4	10
232	C101	Deepak Ragunath Pujari	39	1	PHULENAGAR	7	7	2	7
233	C104	Sachin Inkaram Anne	26	1	PHULENAGAR	4	9	2	9
234	C105	Suhas Inkaram	29	1	PHULENAGAR	4	7	2	7
235	C106	Pradip Aighure Jhagare	60	1	PHULENAGAR	7	0	10	0

The sample size is calculated by the expression,

$$n = \frac{N * Z^2 * p(1-p)}{(N-1)e^2 + Z^2 * p(1-p)}$$

Where,

n = Sample Size

N = Population Size

Z = 1.96

$P = 0.5$

$e = 0.05$

Total Population size	Sample size
235	146

Random sampling was done for each strata using random number generator in MS Excel **"=RANDBETWEEN (lower, upper)"**.

The highlighted rows in Table 1. represents the randomly sampled Households.

1.2. STRATIFIED RANDOM SAMPLING

The population was also sampled using ***Stratified Random Sampling*** technique, where stratification was done based on "**Geographical areas**".

Table 2. Population of Garib Nagar for Stratified Random Sampling

S.No.	HHID	Name of the Head of Household	Area	Household S.No.	Result After Sampling
1	B097	V.D. Jangle	GARIB NAGAR	84	83
2	B096	Prakash Ingle	GARIB NAGAR	56	67
3	B088	Sushant	GARIB NAGAR	90	75
4	B093	Zareena Qureshi	GARIB NAGAR	90	52
5	B094	Prashant Magar	GARIB NAGAR	79	20
6	B045	Manoj Kumar	GARIB NAGAR	76	41
7	B038	Sanjay Shinde	GARIB NAGAR	96	5
8	B035	Santosh Dhotre	GARIB NAGAR	76	30
9	B037	Arun Kamble	GARIB NAGAR	22	71
10	B047	Ravi Bhimrao Khairar	GARIB NAGAR	49	98
11	B049	Pushpa Devi Vishwakarma	GARIB NAGAR	26	7

12	B050	Mangesh	GARIB NAGAR	12	31
13	B051	Reshma	GARIB NAGAR	90	26
14	B053	Kundli Ghughe	GARIB NAGAR	78	69
15	B006	Vinod	GARIB NAGAR	88	71
16	B062	Arjun Nihinat	GARIB NAGAR	32	50
17	B060	Mani	GARIB NAGAR	24	23
18	B059	Chandrakant	GARIB NAGAR	52	11
19	B057	Santosh	GARIB NAGAR	92	31
20	B001	Sudam Raj Guru	GARIB NAGAR	3	11
21	B002	Babban Yadav	GARIB NAGAR	54	42
22	B003	Anuradha	GARIB NAGAR	53	101
23	B005	Pandurang Naik	GARIB NAGAR	77	39
24	B010	Nandakishore Taterao Borde	GARIB NAGAR	60	14
25	B011	Milind	GARIB NAGAR	14	93
26	B012	Mirabai Shinde	GARIB NAGAR	88	52
27	B013	Savita Sirohi	GARIB NAGAR	91	48
28	B014	Dharmendra Vishwakarma	GARIB NAGAR	13	17
29	B015	Abdul Hamid	GARIB NAGAR	86	63
30	B016	Kamla Yadav	GARIB NAGAR	90	15
31	B017	Inamulah Khan	GARIB NAGAR	77	23
32	B018	Harish Sawadekar	GARIB NAGAR	83	52
33	B036	Ram Baran	GARIB NAGAR	38	23
34	B063	Sm Sharma	GARIB NAGAR	27	33
35	B064	Jayshree Nikam	GARIB NAGAR	65	71
36	B065	Babroo Shamrao Gade	GARIB NAGAR	84	31
37	B066	Sanjay Jadhav	GARIB NAGAR	70	35
38	B068	Shyam Lal Pal	GARIB NAGAR	17	19

39	B069	Kavita Somionkar	GARIB NAGAR	57	61
40	B071	Sr Yadav	GARIB NAGAR	37	33
41	B072	Gaya Namdeo Adao	GARIB NAGAR	92	11
42	B073	Jalinder Ubale	GARIB NAGAR	68	19
43	B074	Dashrat Kamble	GARIB NAGAR	86	80
44	B075	Md. SALIM	GARIB NAGAR	70	46
45	B076	Nanda Kakde	GARIB NAGAR	51	58
46	B078	Ramchandra Murdam	GARIB NAGAR	75	94
47	B085	Manoj Kumar Upadhyaya	GARIB NAGAR	48	91
48	B089	Hari Ram	GARIB NAGAR	34	16
49	B092	Sheikh Jahangir	GARIB NAGAR	47	29
50	B025	Noorjahan	GARIB NAGAR	69	48
51	B028	Rahibai Kewde	GARIB NAGAR	52	52
52	B031	Mangal	GARIB NAGAR	33	60
53	B041	Shanti Singh	GARIB NAGAR	41	7
54	B044	Ponraj Nadar	GARIB NAGAR	30	31
55	B055	Rajendra Raikwar	GARIB NAGAR	31	73
56	B056	Sandeep	GARIB NAGAR	14	41
57	B086	Naresh Khandu	GARIB NAGAR	55	36
58	B091	Nasir Ahmed Sheikh	GARIB NAGAR	42	9
59	C036	Ashutosh Gade	GARIB NAGAR	24	74
60	B008	Babasaheb	GARIB NAGAR	13	14
61	B007	Md. Asgar Naddaf	GARIB NAGAR	39	101
62	B024	Md. Azad Khan	GARIB NAGAR	69	16
63	B067	Ramlochan Maurya	GARIB NAGAR	33	35
64	B070	Dayashankar Pandey	GARIB NAGAR	48	56
65	B061	Vinod Gambre	GARIB NAGAR	91	40

66	B039	Ranjit Rathore	GARIB NAGAR	38	97
67	B033	Rajesh Trimbak Nikajhe	GARIB NAGAR	19	18
68	B030	Abdul Rahim Sheikh	GARIB NAGAR	82	32
69	B054	Shoaib Shaikh	GARIB NAGAR	83	44
70	B046	Kalyani	GARIB NAGAR	53	2
71	B042	Aakash Chauhan	GARIB NAGAR	35	34
72	B048	Subadra	GARIB NAGAR	52	11
73	B052	Rajendra Jadhav	GARIB NAGAR	54	99
74	B058	Vijay	GARIB NAGAR	32	36
75	B027	Dhanwantari Parve	GARIB NAGAR	69	61
76	B077	Satish Jadav	GARIB NAGAR	36	79
77	B100	Kijurao	GARIB NAGAR	59	66
78	B099	Rohidas Devgunubale	GARIB NAGAR	69	41
79	B098	Kailash Madhukar Tagatode	GARIB NAGAR	35	3
80	B101	Dinkar Nikalje	GARIB NAGAR	35	91
81	B004	Mahadesh Ishwar Rathore	GARIB NAGAR		
82	B005	Kalyan	GARIB NAGAR		
83	B019	Naunat Kekan	GARIB NAGAR		
84	B020	Kripa Shankar Pandey	GARIB NAGAR		
85	B021	Shawantabai Sitaram	GARIB NAGAR		
86	B022	Ramesh Sable	GARIB NAGAR		
87	B023	Sarjahan	GARIB NAGAR		
88	B026	Asha Krishna Toke	GARIB NAGAR		
89	B029	Vimal Pandagale	GARIB NAGAR		
90	B032	Sunita Kishore Daudve	GARIB NAGAR		
91	B034	Sunil Trimbak Nikaje	GARIB NAGAR		
92	B040	Hari Gadge	GARIB NAGAR		

93	B043	J.R. Mishra	GARIB NAGAR		
94	B079	Surekha	GARIB NAGAR		
95	B080	Dhani	GARIB NAGAR		
96	B081	Santosh Jadhav	GARIB NAGAR		
97	B082	Trimbal Dabhade	GARIB NAGAR		
98	B083	Bhuvneshwar Pathak	GARIB NAGAR		
99	B084	Shantaram	GARIB NAGAR		
100	B087	Ramjitan Prajapati	GARIB NAGAR		
101	B090	Subhan Sheikh	GARIB NAGAR		
102	B095	Leela	GARIB NAGAR		

Table 3. Population of Phule Nagar for Stratified Random Sampling

S.No.	HHID	Name of the Head of Household	Area	Household Serial Number	Result After Sampling
1	C086	Satish	PHULE NAGAR	83	61
2	C107	Suman Sudama Maghade	PHULE NAGAR	32	29
3	C102	Vidyadhar	PHULE NAGAR	51	95
4	C103	Tukaram Shekhba Ingle	PHULE NAGAR	61	106
5	C060	Shyam Bihari Maurya	PHULE NAGAR	28	22
6	C061	Girish Rai	PHULE NAGAR	62	19
7	C062	Urmila Devi	PHULE NAGAR	88	84
8	C063	Brihaspat Chaudhary	PHULE NAGAR	17	55
9	C065	Shobha	PHULE NAGAR	3	77
10	C077	Kesharbai Sukhdev Patekar	PHULE NAGAR	72	105
11	C079	Vilas Uttam	PHULE NAGAR	9	98
12	C080	Sundar Bhishma Jamrao	PHULE NAGAR	89	78

13	C081	Madhukar Sawant	PHULE NAGAR	60	93
14	C082	Sunil Sawant	PHULE NAGAR	103	29
15	C083	Vijaya Baru	PHULE NAGAR	88	101
16	C084	Cristu Raja	PHULE NAGAR	97	75
17	C034	Ravi Umaji Pandit	PHULE NAGAR	68	5
18	C032	Nandlal Gupta	PHULE NAGAR	27	93
19	C030	Ashok B. Janrao	PHULE NAGAR	77	84
20	C028	Jitendra	PHULE NAGAR	85	11
21	C025	Prabhakar More	PHULE NAGAR	86	29
22	C022	Vijay Bahadur Mishra	PHULE NAGAR	83	42
23	C021	Ravi Kishor	PHULE NAGAR	30	67
24	C004	Babu Fakirao Gaekwad	PHULE NAGAR	77	72
25	C026	Balasaheb Gajdhone	PHULE NAGAR	31	41
26	C008	Ranu Narayan Agale	PHULE NAGAR	59	61
27	C003	Dadarao	PHULE NAGAR	100	86
28	C009	Machhindra Waghmare	PHULE NAGAR	49	64
29	C010	Laxman Agale	PHULE NAGAR	13	94
30	C035	Amit Salve	PHULE NAGAR	28	33
31	C039	Dharampal Nadar	PHULE NAGAR	13	87
32	C042	Dayachand Maurya	PHULE NAGAR	24	17
33	C045	Laxman Narayan Bhadarge	PHULE NAGAR	99	6
34	C048	Premchand Nishad	PHULE NAGAR	59	86
35	C051	Neela Phadke	PHULE NAGAR	54	76
36	C053	Baldeep Nirali	PHULE NAGAR	60	67
37	C055	Gorakhnath Lokhande	PHULE NAGAR	26	4
38	C057	Arun Chamatha	PHULE NAGAR	67	104
39	C089	Malati	PHULE NAGAR	45	97

40	C059	Pradeep Chawan	PHULE NAGAR	92	100
41	C092	Vinod Ashok Ingle	PHULE NAGAR	40	83
42	C095	Dhamma Pal	PHULE NAGAR	83	93
43	C097	Kaushalya Gandhle	PHULE NAGAR	58	2
44	C098	Chandrakant Gade	PHULE NAGAR	54	107
45	C099	Umesh Shinde	PHULE NAGAR	73	3
46	C100	Natha Rokade	PHULE NAGAR	1	61
47	C006	Mahesh Kamble	PHULE NAGAR	71	3
48	C013	Anita Singh	PHULE NAGAR	92	4
49	C017	Urmila Vishwakarma	PHULE NAGAR	54	33
50	C020	Premshila	PHULE NAGAR	102	37
51	C023	Sukhdev Sawant	PHULE NAGAR	30	37
52	C038	Ankita Wahude	PHULE NAGAR	96	98
53	C041	Raju Solanki	PHULE NAGAR	42	18
54	C044	Kamal Jadhav	PHULE NAGAR	31	95
55	C047	Kanchan	PHULE NAGAR	41	53
56	C050	Vinay Shinde	PHULE NAGAR	7	38
57	C093	Seema	PHULE NAGAR	79	71
58	C096	Sugandh Rokade	PHULE NAGAR	16	6
59	C108	Nandeo Pandit	PHULE NAGAR	81	38
60	C109	Mohammed Sulaiman Dhulji Chauhan	PHULE NAGAR	43	99
61	C033	Bandu Pandit	PHULE NAGAR	87	23
62	C031	Santosh Patil	PHULE NAGAR	107	94
63	C029	Kamal Ramdev Gawai	PHULE NAGAR	87	35
64	C027	Sukhdev Jadhav	PHULE NAGAR	100	73
65	C024	Mahadev Hazare	PHULE NAGAR	15	95
66	C019	Brijesh Kumar Pandey	PHULE NAGAR	86	33

67	C016	Pradeep Ashok Godhera	PHULE NAGAR	79	29
68	C005	Ambadas Bhimrao Nimbadkar	PHULE NAGAR	61	45
69	C043	Satyadev Rajbhar	PHULE NAGAR	39	53
70	C040	Dilip Padiyar	PHULE NAGAR	72	99
71	C075	Chandu Chalke	PHULE NAGAR	75	83
72	C056	Ganesh Bhadargre	PHULE NAGAR	25	62
73	C068	Hasina Sheikh	PHULE NAGAR	89	52
74	C070	Kamal	PHULE NAGAR	17	70
75	C076	Monica Shantanu Shinde	PHULE NAGAR	7	19
76	C085	Siddharth Chauhan	PHULE NAGAR	108	68
77	C058	Damodar Koli	PHULE NAGAR	31	31
78	C049	Dukni Nishad Malah	PHULE NAGAR	85	33
79	C037	Arjun Shinde	PHULE NAGAR	89	50
80	C046	Santosh Sasare	PHULE NAGAR	60	33
81	C052	Prem Shankar	PHULE NAGAR	33	95
82	C054	Santiam Mishra	PHULE NAGAR	2	85
83	C088	Nagarbai Hazare	PHULE NAGAR	74	85
84	C087	Daya Bai	PHULE NAGAR	30	30
85	C001	Sandhya Zagade	PHULE NAGAR		
86	C002	Bhola Verma	PHULE NAGAR		
87	C007	Anil Sath	PHULE NAGAR		
88	C011	Shahtullah Sheikh	PHULE NAGAR		
89	C012	Harishchandra Maurya	PHULE NAGAR		
90	C015	Thangararaj	PHULE NAGAR		
91	C018	Amar Yadav	PHULE NAGAR		
92	C014	Sushila Devi Hazam	PHULE NAGAR		
93	C064	Ramashrai Pal	PHULE NAGAR		

94	C066	Sundarbai	PHULE NAGAR		
95	C067	Santa Babu Methades	PHULE NAGAR		
96	C069	Shantaram Sakaram Joshi	PHULE NAGAR		
97	C071	Salma Sheikh	PHULE NAGAR		
98	C072	Suman Yadav	PHULE NAGAR		
99	C073	Vithal Chandrapa Kariagowd	PHULE NAGAR		
100	C074	Praveen Saude	PHULE NAGAR		
101	C078	Bala Devi	PHULE NAGAR		
102	C090	Ansaruddin Sheikh	PHULE NAGAR		
103	C091	Arun Sikaram	PHULE NAGAR		
104	C094	Anand Sopah Pargaonbar	PHULE NAGAR		
105	C101	Deepak Ragunath Pujari	PHULE NAGAR		
106	C104	Sachin Inkaram Anne	PHULE NAGAR		
107	C105	Suhas Inkaram	PHULE NAGAR		
108	C106	Pradip Aighure Jhagare	PHULE NAGAR		

Table 4. Population of Ramabai Nagar for Stratified Random Sampling

S.No.	HHID	Name of the Head of Household	Area	Household Serial Number	Result After Sampling
1	A020	Rajanand Pandagare	RAMABAI NAGAR	12	16
2	A015	Padmavati Shirsat	RAMABAI NAGAR	7	9
3	A007	Vithal Laxman Waghmare	RAMABAI NAGAR	2	2
4	A008	Gaurav Sri Sart	RAMABAI NAGAR	1	13
5	A009	Gopal Mathur	RAMABAI NAGAR	24	7
6	A012	Prem Narayan Pandey	RAMABAI NAGAR	13	20
7	A016	Padmavati Pandey	RAMABAI NAGAR	20	19

8	A018	Sushila Ashok Jadhav	RAMABAI NAGAR	16	1
9	A002	Mira Dubey	RAMABAI NAGAR	17	5
10	A003	Kamalabai	RAMABAI NAGAR	21	22
11	A010	Prakash	RAMABAI NAGAR	20	5
12	A014	Muralidhar	RAMABAI NAGAR	5	3
13	A021	Ramesh Pandagare	RAMABAI NAGAR	24	1
14	A022	Kiran Dhediya	RAMABAI NAGAR	13	16
15	A023	Govind Sahni	RAMABAI NAGAR	13	1
16	A011	Sreedhar Vishwanath Kopalwale	RAMABAI NAGAR	12	22
17	A006	Salubai Ramdas Ayera	RAMABAI NAGAR	7	9
18	A004	Hasan Raikbar	RAMABAI NAGAR	23	23
19	A013	Manohar P.	RAMABAI NAGAR	23	17
20	A025	Rajesh Bimajhi	RAMABAI NAGAR	14	16
21	A005	Himmat	RAMABAI NAGAR	4	2
22	A001	Satish Vinkar	RAMABAI NAGAR		
23	A017	Prabash	RAMABAI NAGAR		
24	A019	Aasha	RAMABAI NAGAR		
25	A024	Sanjay Vinkar	RAMABAI NAGAR		

Area	**Total Population Size**	**Sample Size**
Garib Nagar	102	80.4691
Phule Nagar	108	84.3511
Ramabai Nagar	25	21.4723

2. Descriptive Analysis and Visualisation

Table 5. Descriptive Analysis of different Variables

Property	Age of Head of Household	Total Members in each Household	No. of years of Education of HoH
Sum	10712	1160	1643
Mean	45.582979	4.936170213	6.991489362
Mode	40	4	0
Standard Deviation	11.938697	2.056688676	5.968706684
Variance	142.53248	4.229968311	35.62545948

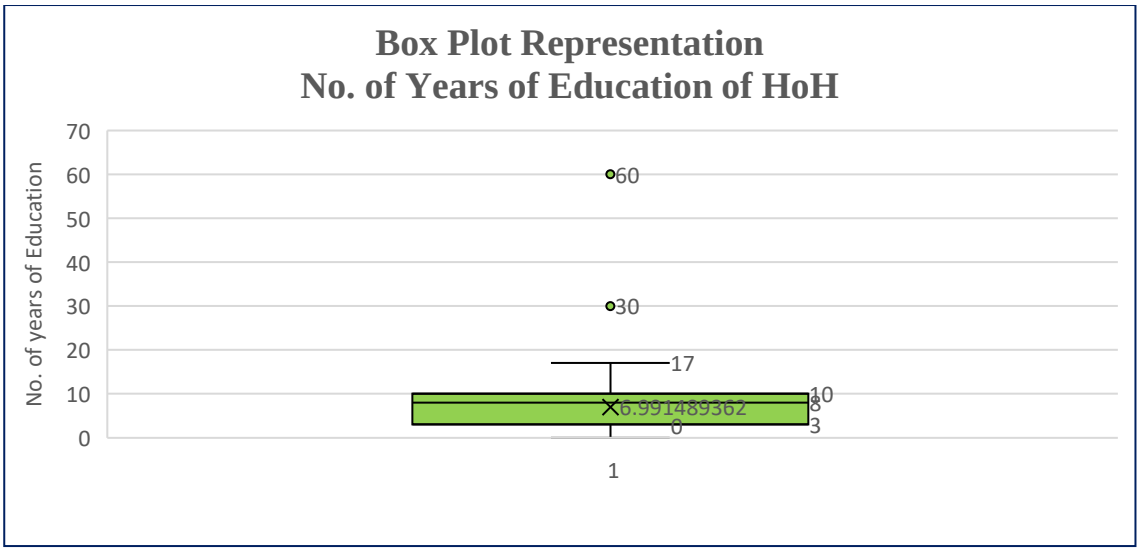
- **Sum:** The total number of years of education across all households (HoH) in the dataset is 10,712.

- **Mean (Average):** The average number of years of education is 45.58 years for the first category, 4.94 years for the second, and 6.99 years for the third. This indicates significant variation in education levels across categories.

- **Mode:** The most frequently occurring value is 40 for the first category, 4 for the second, and 0 for the third.

- **Standard Deviation:** The data shows variability, with the first group having a higher standard deviation (11.94) compared to the others (2.06 and 5.97), indicating more diverse years of education in the first category.

- **Variance:** The variance aligns with the standard deviation, confirming greater dispersion in the first group compared to others.



Graph 1. Box-plot Representation

Key observations include in Graph 1. include:

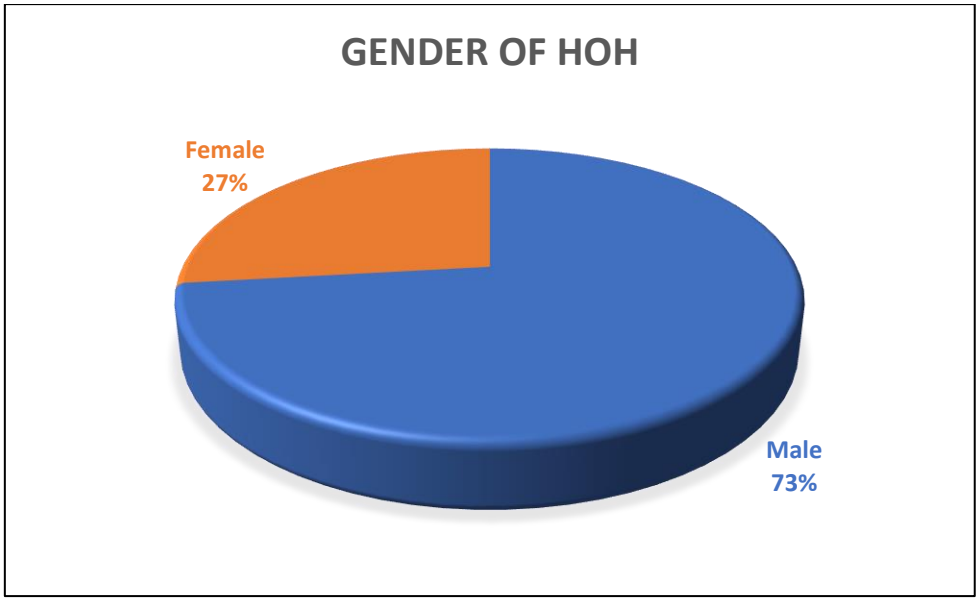
- **Median (Green Line):** The median number of years of education is approximately 7 years, showing that 50% of households have education levels below this value.

- **Interquartile Range (IQR):** The data is concentrated between 3 years (Q1) and 10 years (Q3) of education. This narrow range highlights that most households have low to moderate education levels.

- **Whiskers:** The lower whisker extends to 0 years, showing households with no education. The upper whisker extends to 17 years, indicating the maximum range of typical education levels.

- **Outliers:** Outliers are present at 30 years and 60 years, suggesting a small number of highly educated heads of households that are exceptions to the general trend.

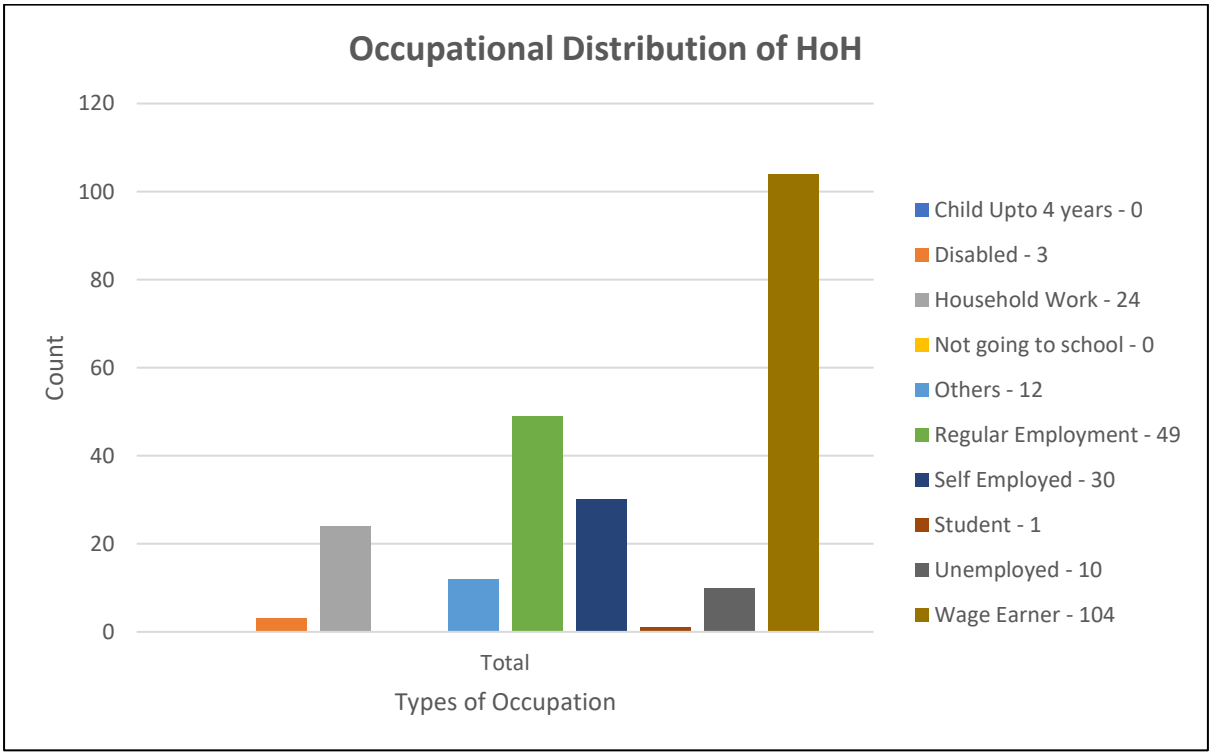
Male	172
Female	63



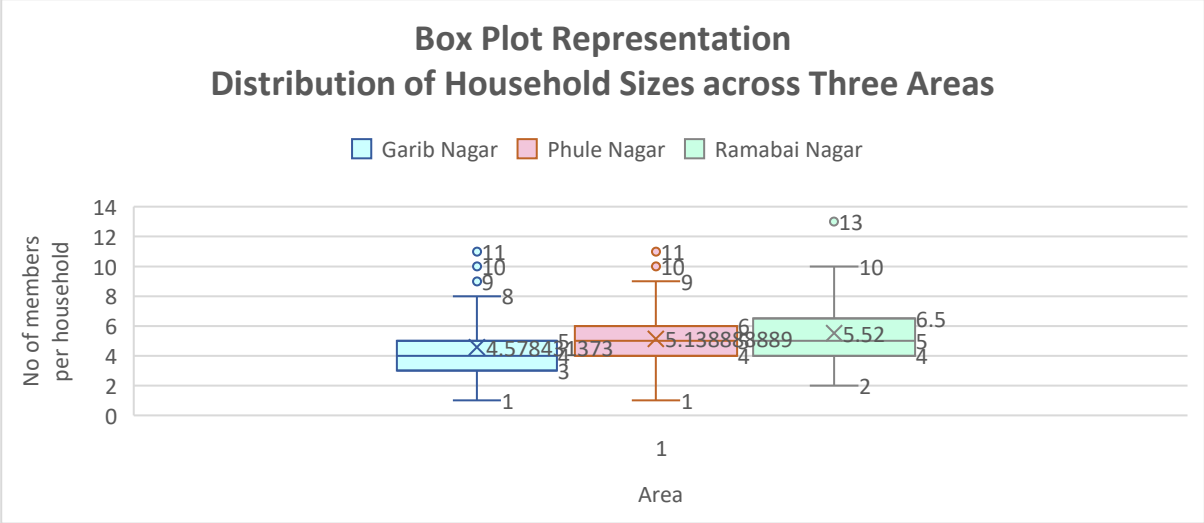
Graph 2. 3D-Pie Chart representation

Table 6. Occupation-Count data of Head of Household

Code	Criteria (Occupation)	Count
1	Household Work	24
2	Wage Earner	104
3	Regular Employment	49
4	Self Employed	30
5	Student	1
6	Unemployed	10
7	Child Upto 4 years	0
8	Not going to school	0
9	Disabled	3
10	Others	12



Graph 3. Pivot Chart representation



Graph 4. Box-plot representation

i. **Garib Nagar:**

- **Median Household Size**: Approximately 4.57 members.
- **Interquartile Range (IQR)**: From 3 to 5 members.
- **Range**: 1 to 8 members.
- **Outliers**: Two outliers are observed (household sizes of 10 and 11 members).

ii. **Phule Nagar:**

- **Median Household Size**: Approximately 5.13 members.
- **IQR**: From 4 to 9 members.
- **Range**: 1 to 11 members.
- **Outliers**: One outlier is observed (11 members).

iii. **Ramabai Nagar:**

- **Median Household Size**: Approximately 5.52 members.
- **IQR**: From 4 to 6.5 members.
- **Range**: 2 to 10 members.
- **Outliers**: One outlier is observed (household size of 13 members).

iv. **Comparative Insights:**

Ramabai Nagar has the highest median household size (5.52), followed by Phule Nagar (5.13), and Garib Nagar (4.57). Phule Nagar shows the greatest variability, as indicated by a wider IQR and range. All areas show the presence of outliers, indicating some unusually large households.

3. Hypothesis Testing and Inferential Analysis

3.1. HYPOTHESIS

3.1.1. Null Hypothesis (H_0): *The mean length of residential history is the same for homeowners and non-homeowners.*

3.1.2. **Alternative Hypothesis (H_1):** *The mean length of residential history differs between homeowners and non-homeowners.*

Table 7. Anova: Single Factor

SUMMARY

Groups	*Count*	*Sum*	*Average*	*Variance*
Residential Years for homeowners	192	6857	35.71354	159.6714
Residential Years for non-homeowners	43	1009	23.46512	299.3976

ANOVA

Source of Variation	*SS*	*df*	*MS*	*F*	*P-value*	*F crit*
Between Groups	5270.628	1	5270.628	28.51175	2.2095E-07	3.881679
Within Groups	43071.94	233	184.8581			
Total	48342.57	234				

3.2. KEY RESULTS

3.2.1. F-statistic (F): The calculated F-value is 28.51175.

3.2.2. P-value: The P-value is 2.2095E-07 (very close to 0).

3.2.3. Critical F-value (F-crit): The critical F-value is 3.881679 at the chosen significance level.

3.3. INTERPRETATION

The P-value is ***much smaller*** than the common significance level (0.05). This indicates that the differences observed are statistically significant. The F-statistic is ***larger*** than the critical F-value, further supporting that the differences between groups are not due to random chance. Since the P-value is ***extremely small*** and the F-statistic ***exceeds the critical value***, we ***reject the null hypothesis (H_0).***

This means there is enough evidence to conclude that the mean length of residential history differs between homeowners and non-homeowners. Homeowners tend to have a longer residential history compared to non-homeowners.

4. Correlation and Regression Analysis

We selected "ownership status" and "residential history" from the given dataset for correlation and regression analysis.

4.1. CORRELATION

Correlation Coefficient is **-0.330192**, this indicates a moderate negative correlation between ownership status and residential history. As the length of residential history increases, the likelihood of housing ownership tends to decrease.

4.2. REGRESSION ANALYSIS

Null Hypothesis (H_0): *Ownership of housing is not associated with the length of residential history in the region.*

Alternate Hypothesis (H_1): *Ownership of housing is more prevalent among households with a longer residential history.*

Table 8. Regression Analysis Table

Regression Statistics	
Multiple R	0.33019182
R Square	0.10902663
Adjusted R Square	0.10520271
Standard Error	0.36652644
Observations	235

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	1	3.830314454	3.830314	28.51174557	2.20948E-07
Residual	233	31.30160044	0.134342		
Total	234	35.13191489			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*	*Lower 95.0%*	*Upper 95.0%*
Intercept	1.48092532	0.060705819	24.39511	4.2E-66	1.361322859	1.60052778	1.36132286	1.60052778
From when are you living in Mumbai (in years)	-0.00890128	0.001667019	-5.33964	-2.20948E-07	-0.012185634	-0.0056169	-0.01218563	-0.0056169

4.2.1. Regression Equation:

The relationship between ownership status (dependent variable i.e. **y**) and residential history (independent variable i.e. **x**) can be expressed as:

$$y = 1.480925 - 0.008901 \times x$$

$$Intercept = 1.480925$$

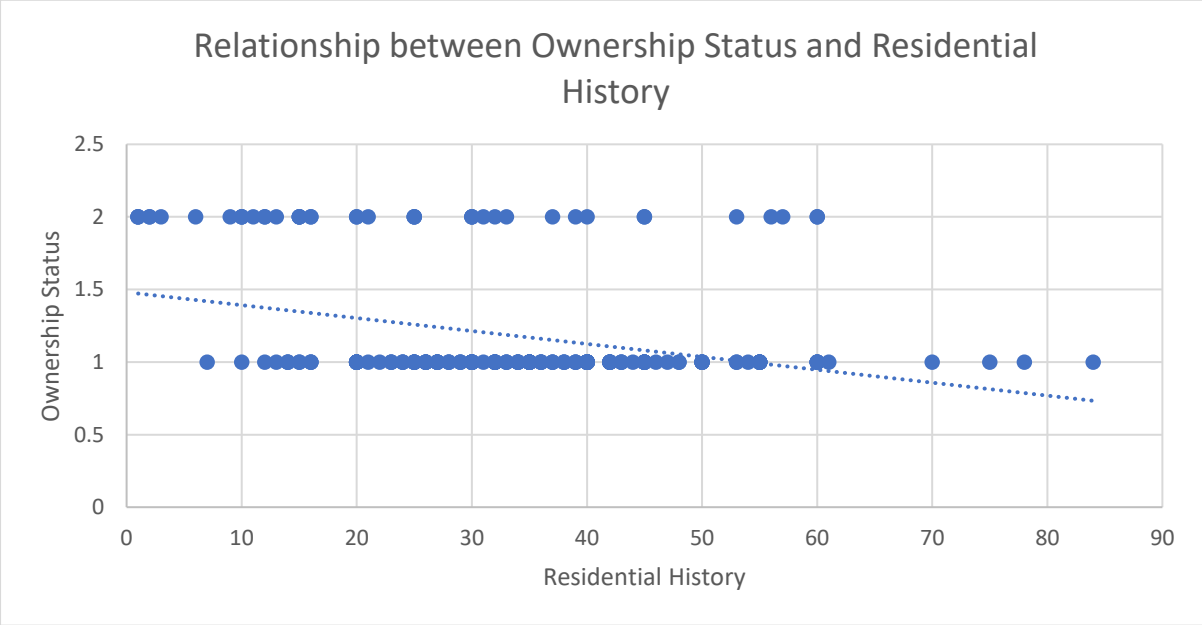
$$Slope = -0.008901$$

4.2.2 Key Outputs:

i. **R^2 (R-Square)**: 0.109027. Only 10.9% of the variability in ownership status is explained by residential history. This suggests that other factors might also significantly impact ownership status.

ii. **F-statistic (28.512)**: The model is statistically significant, as the P-value (2.209E-07) is far below 0.05. This means the relationship between ownership status and residential history is unlikely due to random chance.

iii. **T-statistic for Slope (-5.34)**: The negative slope is statistically significant, with a very low P-value (2E-07). This reinforces that residential history has a meaningful (though small) impact on ownership status.



Graph 5. Regression curve

The scatterplot shows a ***weak downward trend*** in ownership status as residential history increases, aligning with the ***negative correlation*** and ***regression results***. There is a ***moderate negative relationship*** between residential history and ownership status. Longer residential history slightly reduces the likelihood of housing ownership. The regression model is ***statistically significant*** but explains only a small portion of the variation, indicating other factors also influence housing ownership.

5. Conclusion

This was our attempt at analysing the raw dataset using the statistical methods learnt over the coursework of ***Research Methods II (Quantitative).*** We looked at the data from multiple viewpoints and finally concluded on the factors we found to be important. This study employed statistical methods to explore patterns and relationships within the provided dataset, with a focus on demographics, education, household size, and housing ownership. The analysis revealed key insights into the socio-economic dynamics of the surveyed population:

5.1. Descriptive Findings: The descriptive analysis highlighted notable variations in the years of education among household heads, with a significant portion having low to moderate education levels. The median household sizes across different areas demonstrated variability, with Ramabai Nagar showing the highest median and Phule Nagar the greatest variability.

5.2. Hypothesis Testing: The ANOVA results established a statistically significant difference in the residential history of homeowners and non-homeowners, rejecting the null hypothesis. Homeowners were found to have longer residential histories, indicating stability as a potential factor in ownership patterns.

5.3. Correlation and Regression Analysis: A moderate negative correlation between residential history and housing ownership was identified. The regression model, while statistically significant, explained only a small fraction of the variability, suggesting that additional factors influence ownership status.

Overall, the findings provide a nuanced understanding of the interplay between residential history, education, and household characteristics. However, the low explanatory power of the regression model indicates the need for further research incorporating additional variables such as income levels, employment stability, and access to housing resources. These results can inform targeted interventions and policies aimed at enhancing housing accessibility and addressing socio-economic disparities.